

UNIT III – LASERS & FIBER OPTICS

Lasers - Principles of Laser – Spontaneous and Stimulated Emissions - Einstein's Coefficients – Population Inversion and Laser Action – types of Optical resonators (qualitative ideas) – Types of Lasers - NdYAG, CO₂ laser, GaAs Laser-applications of lasers

Fiber Optics - Principle and Propagation of light in optical fiber – Numerical aperture and acceptance angle – Types of optical fibers (material, refractive index, mode) - applications to sensors and Fibre Optic Communication

UNIT III – LASERS & FIBER OPTICS

INTRODUCTION

We know current can be amplified by vacuum tube or transistor amplifier. Similarly light waves can also be amplified and is termed as 'LASER' which is an acronym for "Light Amplification by Stimulated Emission of Radiation". After the discovery of laser by T.H.Maiman in 1960, a rigorous analysis were made, which led to many types of laser got due to the lasing action with atoms, ions molecules etc., in gases, solids and liquids. The light beam from laser has frequency upto 10^{14} Hz lead to many application in the scientific world and showed that the function of optics is alive.

3.1 Characteristics of Laser

Laser is basically a light source and it has the following important properties

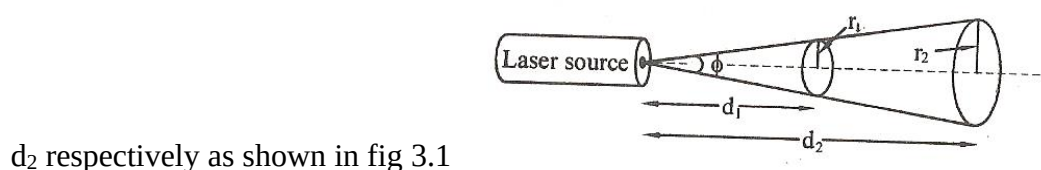
- (i) High Directionality
- (ii) High Intensity
- (iii) High Monochromatic
- (iv) High Coherence

1. Directionality:

Ordinary light spreads in all directions and its angular spread is 1 metre/metre. But it is found that laser is highly directional and its angular spread is 1mm/metre. For example the laser beam can be focused to very long distance with a low divergence or angular spread.

Divergence (or) Angular spread is given by $\phi = \frac{r_2 - r_1}{d_2 - d_1}$

Where d_1, d_2 are two distances from the laser source r_1, r_2 are the radii of the beams spot at a distance d_1 and



d_2 respectively as shown in fig 3.1

Fig 3.1 Directionality

2. Intensity:

Since an ordinary light spreads in all direction, the intensity reaching a distant target is very less. But in the case of laser, due to high directionality the intensity of laser beam reaching the target is of high intensity . For example, 1 milliwatt power of He-Ne laser appears to be brighter than the sunlight.

3. Monochromatic:

Laser beam is highly monochromatic i.e., the wavelength is single, whereas in ordinary light like mercury vapour lamp, many wavelengths of light are emitted.

4. Coherence:

It is an important characteristic of laser beam. In laser the wave trains of same frequency are in phase i.e. the radiation given out is in mutual agreement not only in phase but also in the direction of emission and polarization. Thus, it is a coherent beam. Due to high coherence it results in an extremely high power density.

Due to these peculiar properties of laser, it has wide engineering and medical applications.

3.2 Difference between ordinary light and laser beam.

S. No.	Ordinary light	Laser light
1.	In ordinary light the angular spread is more.	In laser beam the angular spread is less.
2.	They are not directional.	They are highly directional.
3.	It is less intense	It is highly intense.
4.	It is not a coherent beam and is not in phase.	It is a coherent beam and is in phase.
5.	The radiations are polychromatic.	The radiations are monochromatic

3.3 EINSTEIN'S QUANTUM THEORY OF RADIATION

Determination of Einstein's coefficients (A & B)

We know that, when light is absorbed by atoms (or) molecules, then they are excited from the lower energy level (E_1) to the higher energy level (E_2) and during the transition from higher energy level (E_2) to lower energy level (E_1) the light is emitted from the atoms (or) molecules.

According to Einstein it is impossible to predict which atom is going to be transition from lower energy level (E_1) to the higher energy level (E_2). But it is possible to calculate the rate of transition by using the law of probability.

When an atom is exposed to (light) photons of energy $E_2 - E_1 = h\nu$, three distinct process take place.

- (i) Absorption
- (ii) Spontaneous emission
- (iii) Stimulated emission

(i) Absorption:

An atom in the lower energy level or ground state energy level E_1 absorbs the incident photon radiation of energy $h\nu$ and goes to the higher energy level (or) excited energy state E_2 as shown in fig 3.2. This process is called as **absorption**.

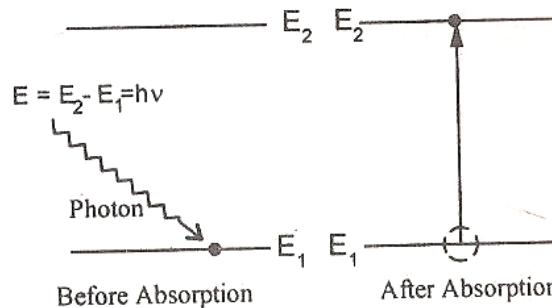


Fig 3.2 Absorption

If there are many number of atoms in the ground state then each atom will absorb the energy from the incident photon and goes to the excited state then, the rate of absorption R_{12} is proportional to the following factors

i.e. $R_{12} \propto$ energy density of incident radiation (ρ_ν)

$R_{12} \propto$ No : of atoms in the ground state N_1

$$\text{i.e. } R_{12} \propto \rho_\nu N_1 \dots\dots\dots (1)$$

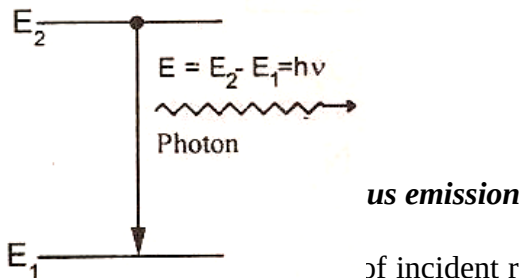
$$(\text{or}) R_{12} = B_{12} \rho_\nu N_1$$

Where B_{12} is a constant which gives the probability of absorption transition per unit time. Normally, the atom in the excited state will not stay there for a long time, rather it comes to ground

state by emitting a photon of energy $E=h\nu$ such an emission takes place by one of the following two methods.

(ii) Spontaneous emission:

The atom in the excited state returns to the ground state by emitting a photon of energy $E= E_2-E_1=h\nu$, spontaneously without any external triggering as shown in fig 3.3. This process is known as **spontaneous emission**.



Such an emission is called **spontaneous emission**. If N_1 and N_2 are the no. of atoms in the ground state (E_1) and excited state (E_2) respectively, then the rate of spontaneous emission is

$$R_{21} \text{ (SP)} \propto N_2$$

$$R_{21} \text{ (SP)} = A_{21}N_2 \dots\dots\dots (2)$$

(iii) Stimulated emission:

The atom in the excited state can also return to the ground state by external triggering (or) inducement of photon thereby emitting a photon of energy equal to the energy of the incident photon, known as **stimulated emission**. Thus results in two photons of same energy, phase difference and of same directionality as shown in fig. 3.4.

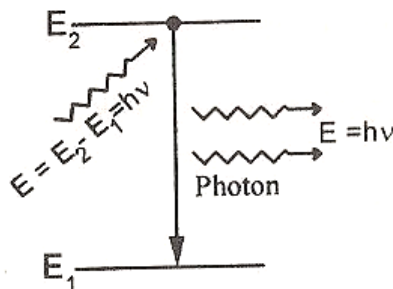


Fig. 3.4 Stimulated Emission

The rate of stimulated emission is given by

$$R_{21} \propto \rho_{\nu} N_2$$

$$R_{21} = B_{21} \rho_{\nu} N_2 \quad \dots (3)$$

Where B_{21} is a constant which gives the probability of stimulated emission transition per unit time.

Expressions for the coefficients A_{21} , B_{12} and B_{21}

Einstein's theory of absorption and emission of light by an atom is based on Planck's theory of radiation. Also under thermal equilibrium, the population of energy levels obey the Boltzmann's distribution law.

i.e. Under thermal equilibrium

The rate of absorption = The rate of emission

i.e. $\text{Eq}(1) = \text{Eq}(2) + \text{Eq}(3)$

$$B_{12} \rho_{\nu} N_1 = A_{21} N_2 + B_{21} \rho_{\nu} N_2 \quad \dots (1)$$

$$\rho_{\nu} (B_{12} N_1 - B_{21} N_2) = A_{21} N_2 \quad \dots (2)$$

$$\rho_{\nu} = \frac{A_{21} N_2}{B_{12} N_1 - B_{21} N_2} \quad \dots (3)$$

Dividing the above equation both on Numerator and Denominator by N_2

$$\rho_{\nu} = \frac{A_{21}}{B_{12} \left(\frac{N_1}{N_2} \right) - B_{21}} \quad \dots (4)$$

We know from Boltzmann's distribution law

$$N_1 = N_0 e^{(-E_1/K_B T)}$$

$$\text{Similarly } N_2 = N_0 e^{(-E_2/K_B T)}$$

Where, K_B is the Boltzmann's constant

T is the absolute temperature and

N_0 is the number of atoms at absolute zero.

At equilibrium, we can write the ratio of population levels as follows.

i.e.

$$\frac{N_1}{N_2} = e^{(E_2 - E_1)/K_B T}$$

Since $E_2 - E_1 = h\nu$, we have

$$\frac{N_1}{N_2} = e^{h\nu / K_B T} \dots\dots\dots (5)$$

Substituting eqn (5) in eqn (4) we have

$$\rho_\nu = \frac{A_{21}}{B_{12}(e^{h\nu / K_B T}) - B_{21}}$$

Divide the above equation both on Numerator and Denominator by B_{21}

$$\rho_\nu = \frac{A_{21}}{B_{21}} \frac{1}{\left[\frac{B_{12}}{B_{21}} e^{(h\nu / K_B T)} - 1 \right]} \dots\dots\dots (6)$$

This equation has very good agreement with Planck's energy distribution radiation law.

$$\rho_\nu = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{(h\nu / K_B T)} - 1} \dots\dots\dots (7)$$

Therefore comparing eqn(6) and eqn(7) we can write $B_{12} = B_{21} = B \dots\dots (8)$

and

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3} \dots\dots\dots (9)$$

Taking $A_{21} = A$ and the constants A and B are called as **Einstein Coefficients**.

RESULT:

- (i) Equation (8) states that the stimulated emission rate per atom is the same as the absorption rate per atom.
- (ii) Equation (9) gives the relation between the spontaneous emission and stimulated emission coefficients. Since the ratio is proportional to γ^3 , the probability of spontaneous emission rapidly increases with the energy differences between the two states.

3.4 Requirements of Metastable states:

Eqn (9) further indicates that stimulated transitions will dominate the spontaneous transitions if the value of the ratio B_{12}/A_{21} is large.

To increase the probability of stimulated emissions, the life time of atoms at the excited state should be large. In other words, it is necessary that the excited state has a longer life time. A metastable state is such a state. Atoms excited to the metastable state remain excited for an appreciable

time, which is of the order of 10^{-6} to 10^{-3} sec, this is 10^3 to 10^6 times the life times of the ordinary excited energy levels. It means that the active centres must have metastable states.

3.5 PRINCIPLE OF LASER ACTION

Let us consider many number of atoms in the excited state. We know that photons emitted during stimulated emission have same frequency, energy and are in phase as the incident photon. This results in 2 photons of similar properties. These two photons indicate stimulated emission of 2 atoms in excited state thereby resulting in 4 photons. These 4 photons induce 4 more atoms and give rise to 8 photons. etc., as shown in fig 3.5

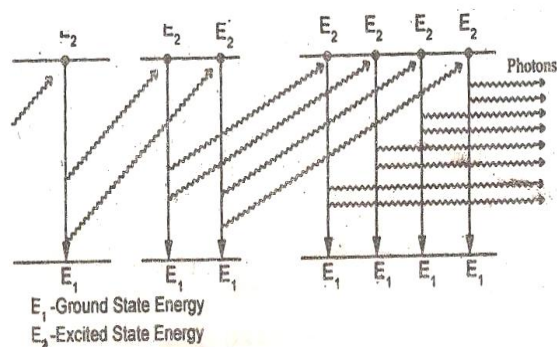


Fig 3.5 Lasing Action

Due to stimulated emission the photons multiply in each step giving rise to an intense beam of photons that are coherent and moving in the same direction. Hence the light is amplified by stimulated emission of radiation termed as LASER.

3.6 Components of Laser

Laser system has three important components. They are:

- (i) Active medium.
- (ii) Pumping source.
- (iii) Optical resonator.

(i) Active medium:

It is the basic essential material in which atomic transition takes place, leading to laser action. Active Laser media are available in all aggregate state (i.e.) solid, liquid & gaseous state.

(ii) Pumping source:

It is the device with which population inversion can be achieved in the active medium.

(iii) Optical resonator:

- The optical resonator constitutes an active medium kept in between a 100% reflecting mirror and a partially reflecting mirror.
- This optical resonator acts a feedback system in amplifying the light emitted from the active medium, by making it to undergo multiple reflections between the 100% mirror and the partial mirror.

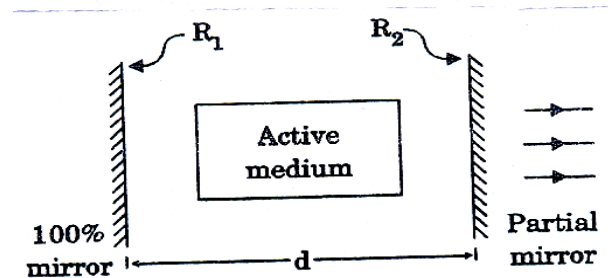


Fig. 3.6 Optical Resonator

3.7 POPULATION INVERSION :

The state of achieving more number of atoms in the excited state (N_2) than the ground state (N_1) is called as **population inversion**. $N_2 > N_1$.

Consider two energy level systems E_2 and E_1 representing upper and lower state .We know that when a photon of energy equal to the differences in energy between two levels approaches the two atom, both the stimulated emission and absorption process are equally possible.

Under thermal equilibrium the population of the energy levels obey Boltzmann's distribution given by the equation.

$$N_1 = \exp [-(E_2 - E_1)/kT]$$

Let N_1 be the number of atoms in ground state

N_2 be the number of atoms in excited state. Then,

For stimulated emission to dominate, it is necessary to increases the population of the upper energy level so that this is greater than lower level. This is known as “**population inversion**”. The material having a population inversion is called “**active material**”

Condition for population inversion:

- (i) There must be atleast a pair of energy levels.
- (ii) There must be a source to supply energy to the medium.
- (ii) The atoms must be raised continuously to the excite state.

3.8 Pumping method:

The process of raising more number of atoms to excited state by means of an intense external source is called as **pumping process**.

There are several methods by which the population inversion (pumping) can be achieved. Some of the most commonly used methods are

- a) Optical pumping
- b) Direct electron excitation (Electric discharge)
- c) Inelastic atom-atom collision
- d) Direct conversion
- e) Chemical process

a) Optical pumping:

Here the atoms are excited with the help of photons emitted by an external optical source. The atoms absorb energy from the photons and raise to excited state. (e.g.) Ruby laser, Nd-YAG laser.

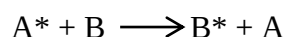
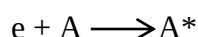
b) Direct electron excitation:

The electrons are accelerated to very high velocities by strong electric field and they collide with gas atoms and these atoms are raised to excited state (e.g.) Argon laser

c) Inelastic atom-atom collision:

In this method a combination of two types of gases are used say A and B both having same or nearly coinciding excited states A* and B*

During electric discharge A atoms get excited due to collision with electron. The excited A* atom now collides with B atoms so that B goes to excited state B* e.g. He-Ne laser, CO₂ laser.



d) Direct conversion:

Due to electrical energy applied in direct band gap semiconductor like GaAs etc. The combination of electrons and holes takes place and electrical energy is converted into light energy directly. E.g. semiconductor laser.

e) Chemical method:

Due to some chemical reactions, the atoms may be raised to excited state (e.g.) Dye laser

3.9 OPTICAL RESONATOR:

Laser is a light source and it is analogous to an electronic oscillator. An electronic oscillator is essentially an amplifier supplied with positive feedback. A part of the output of the amplifier is taken and fed back at its input.

When the amplifier is switched on, electrical noise signal of appropriate frequency present at input will be amplified the output is fed back to the input and amplified again and so on. A stable output is quickly reached when the oscillator acts as a source of a particular frequency.

In laser the active medium is the amplifier which is converted into an oscillator through the feedback mechanism established by an optical resonator. A pair of optically plane parallel mirrors, enclosing laser medium in between them is known as an **optical resonant cavity** which is shown in Fig. 3.7.

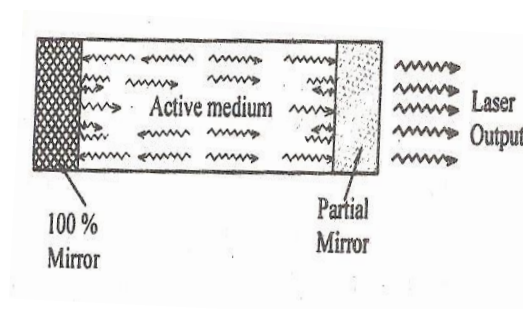


Fig. 3.7 Optical Resonator

One of these mirrors is partially reflecting and the other is made fully reflecting. In laser, the role of noise is played by photons emitted spontaneously. The photons emitted along the optic axis of the resonant cavity travel through the medium and trigger stimulated emission. They are reflected by the end mirror and reverse their path.

The photons are fed back into the medium and travel toward the opposite end mirror causing more stimulated emissions. The photons are once more reflected at the mirror and travel toward the opposite mirror. Substantial light amplification takes place because the light beam is reflected several times at the mirrors and gains strength in each passage ultimately when the amplification balances the losses in the cavity the laser beam emerges out from the front end mirror. In the absence of resonator cavity, there would be no amplification of light.

3.10 TYPES OF LASERS

Based on the type of active medium, Laser systems are broadly classified into the following category

Sl. No.	Types of Laser	Examples
1.	Solid State Laser	Ruby Laser, Nd-YAG Laser
2.	Gas Laser	He-Ne Laser, Co ₂ Laser, Argon-ion laser
3.	Liquid Laser	Europium Chelate Laser
4.	Dye Laser	Rhodamine 6G Laser, Coumarin dye Laser
5.	Semi conductor Laser	GaAs Laser, GaAsP Laser

3.10.1 Nd-YAG (NEODYMIUM – YTTRIUM ALUMINIUM GARNET LASER)

Characteristics of Nd- YAG Laser

1. Type: Doped insulator laser (solid state laser)
2. Active medium: Yttrium Aluminium garnet ($\text{Y}_3\text{Al}_5\text{O}_{12}$)
3. Active medium: Neodymium (Nd^{3+} ions)
4. Pumping method: Optical pumping
5. Pumping Source: Xenon flash lamp
6. Optical Resonator: Ends of the rods polished and silvered and two mirrors one of which is totally reflecting and the other partially reflecting.
7. Power output: 2×10^4 Watts
8. Wavelength emitted: $1.064 \mu\text{m}$.

Introduction:

Nd-YAG laser is a doped insulator laser. It is a four level system in which the active medium is taken in the form of a crystal. Here the crystal is intentionally doped during its growth. Those types of lasers have number of energy levels with same energy. The laser is used to generate high power intensity.

Principle:

The term “Doped Insulator Laser” refers to the active medium, Yttrium Aluminium Garnet doped with Neodymium Nd^{3+} ions has many energy levels. Due to optical pumping these ions are raised to excited levels. During the transition from metastable state to E_1 state, the laser beam of wavelength $1.064 \mu\text{m}$.

Construction:

The active medium is made as a rod which has Yttrium aluminium garnet ($\text{Y}_3\text{Al}_5\text{O}_{12}$) doped with a rare earth metal ion Neodymium Nd^{3+} ions . The Nd^{3+} ion normally occupies the Yttrium ions and provides the energy levels for both lasing transitions and pumping. This rod is placed inside a highly reflecting elliptical cavity as shown in fig 3.8

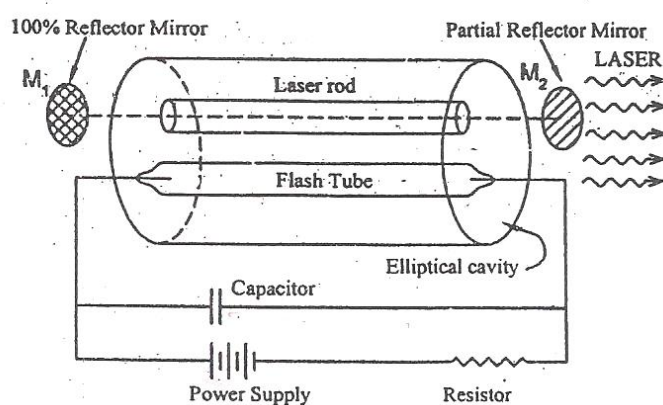


Fig. 3.8 Nd – YAG Laser

A close optical coupling is made by placing the Xenon flash lamp near by the laser rod, in such a way that most of the radiations from the flash tube passes through the laser rod due to the elliptical cavity. The flash tube may be switched ON and controlled with the help of a capacitor. The discharge of capacitor is initiated using a high voltage source. The optical resonator is formed by grinding the ends of the rods and coated with silver accompanied by two mirrors, one is 100% reflecting and the other is partially reflecting which, is included to increase the efficiency of the output beam.

Working:

1. The xenon flash lamp is switched ON and the light is allowed to fall on the laser rod.
2. The intense white light excites the Neodymium (Nd^{3+}) ions from the ground state to various energy levels above E_2 . Hence the atoms are raised to group of higher levels in E_3 as shown in Fig. 3.9 in the energy level diagram.

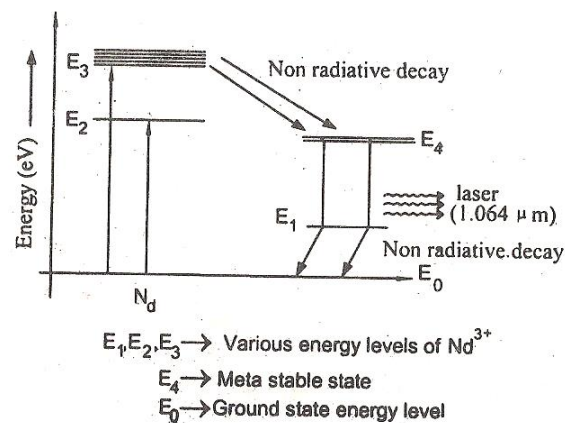


Fig. 3.9 Energy level of Nd-YAG Laser

3. From these energy levels the ions make non-radioactive decay and are gathered in a state called as Metastable state, until the population inversion is achieved.
4. Once the population inversion is achieved, the stimulated is achieved.
5. Hence pulses of laser beam of wavelength $1.064 \mu\text{m}$ is emitted during the transition from E_4 to E_1 (lower)
6. A large amount of heat is produced by the flash tube during the working. Hence cooling arrangement is made either by blowing air (or) circulating water over the crystal.

Applications of Nd-YAG Laser:

1. It is used in transmitting signals to longer distances.
2. It is used in long haul communication system.
3. It is also used in the endoscopic applications.
4. It plays a vital role in remote sensing applications.

3.10.2 CARBON-DI-OXIDE (CO₂) LASER

Characteristics of CO₂ laser

1. Type: Gas Laser
2. Active medium: Mixture of CO₂, N₂ and Helium (or) Water vapour.
3. Active centre: CO₂
4. Pumping method: Electric discharge method
5. Optical Resonator: Metallic mirror of gold (or) silicon mirrors coated with Aluminium.
6. Power output: 10 KW
7. Nature of output: continuous (or) pulsed
8. Wavelength of the output: 9.6 μm & 10.6 μm

Principle:

The transition between these vibrational and rotational energy levels leads to the construction of molecular gas laser. Here the Nitrogen atoms are initially raised to excited state. The nitrogen atoms deliver the energy to CO₂ atoms which has closest energy level to it. Then, transition takes place between the vibrational energy levels of the CO₂ atoms and hence laser beam is emitted.

The molecular gas lasers have two types of transitions.

- (1) Transitions between vibrational states of the same electronic state (Fig. 3.10.).
- (2) Transitions between vibration levels of different electronic state (Fig. 3.11)

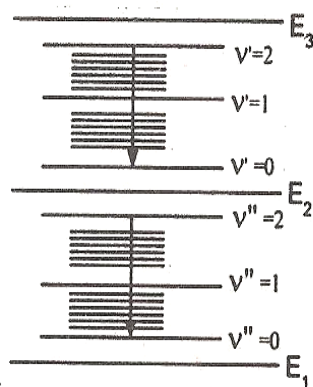


Fig. 3.10

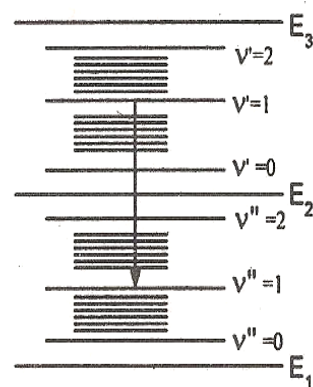


Fig. 3.11

CO₂ laser satisfies the first condition, here the laser transition occurs between vibrational energy levels of the same electronic state.

Fundamental modes of vibration of CO₂ molecule

(i) Symmetric stretching mode(10°0)

Here the carbon atom is stationary and the oxygen atoms oscillate or vibrate along the axis of the molecule as shown in fig. 3.12

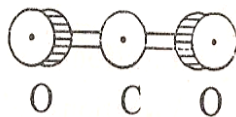


Fig. 3.12 Symmetric mode

The two oxygen atom simultaneously approaches or departs with respect to carbon atom. The state of vibration is given by 3 integers (m n l q) here $\underline{10^00}$ which corresponds to the degree of excitation.

(ii) Asymmetric stretching mode (00°1), (00°2)

Here all the three atoms will vibrate. Here the oxygen atoms vibrate in the opposite direction to the vibration of carbon atom as shown in fig. 3.13

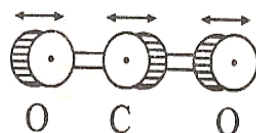


Fig. 3.13 Asymmetric mode

It gives the quanta of frequency (00°1), (00°2)

(iii) Bending mode (01°0, 02°0)

Here the atoms will not be linear rather the atoms will vibrate perpendicular to the molecular axis as shown in fig. 3.14 This gives rise to two quanta of frequency represented by (01°0, 02°0)

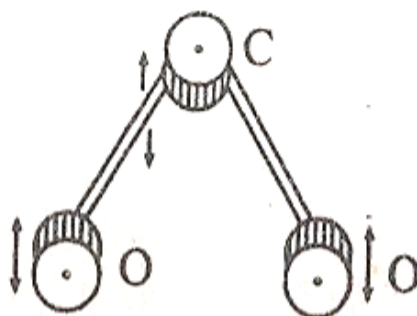


Fig 3.14 Bending mode

Construction:

It consists of a discharge tube in which CO₂ is taken along with nitrogen and helium gases with their pressure of 0.33:1.2:7mm of Hg for CO₂, nitrogen and the He respectively. Nitrogen helps to increase the population of atoms in the upper level of CO₂. While Helium helps to depopulate the atoms in the lower level of CO₂ and also to cool the discharge tube. The discharge is produced by D.C excitation. At the ends of the tube sodium chloride/Brewster windows are placed as shown in fig 3.15.

Confocal silicon mirrors coated with aluminum or metallic mirror of gold is employed for proper reflection, which form the resonant cavity. The output power can be increased by increasing the diameter of the tube.

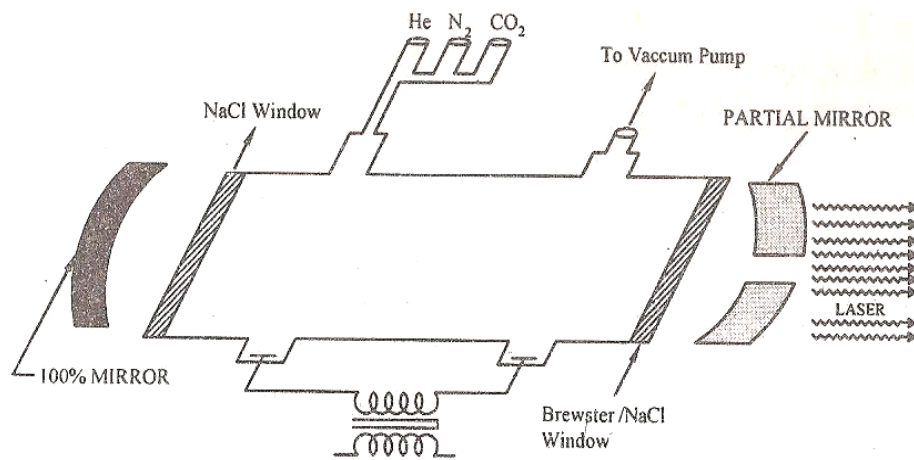
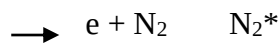


Fig. 3.15 CO₂ Laser

Working:

1. The discharge is passed through the tube first the nitrogen atoms are raised to excited state.



2. The excited N₂ atoms undergo resonant energy transfer with CO₂ atom and raises CO₂ (00°1) to excited state due to closer energy level CO₂ (00°1) and Nitrogen



3. When transition takes place between 00°1 to 10°0, laser of wavelength 10.6μm is emitted as shown in fig 3.16
4. Similarly, when transition takes place between 00°1 and 02°0 laser beam of wavelength 9.6 μm is emitted as shown below.
5. Since 00°1 → 10°0 has a higher gain than 00°1 → 02°0 transitions. Usually the laser beam of wavelength 10.6 μm is produced more.
6. When the gas flow is longitudinal power output is 50 to 60 watts but if the gas flow is perpendicular to the discharge tube the output power may be raised to 10 Kw/m.

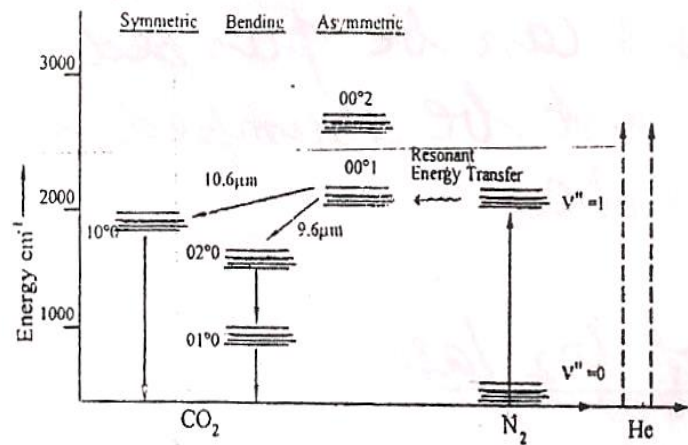


Fig. 3.16 Energy Level Diagram

7. This type of CO₂ laser is known as TEA laser (Transversely Excited Atmospheric Pressure Laser)
8. The combination of carbon monoxide and oxygen will also have some effect on the laser action. To avoid these unused gases can be pumped out and fresh CO₂ must be pumped inside the discharge tube.

Application of CO₂ Laser

1. This laser has applications in medical field such as neurosurgery, treatment of liver, lungs and also is bloodless operations.
2. It is widely used in open air communications.
3. This layer also has wide application over military field.

3.10.3 HOMOJUNCTION SEMICONDUCTOR LASER (GaAs LASER)

Characteristics:

1. Type: Homo junction semiconductor laser
2. Active medium: P-N junction diode.
3. Active Centre: Recombination of electrons and holes.
4. Pumping: Direct pumping.
5. Optical Resonator: Junction of diodes- polished.
6. Power output: 1 mW
7. Nature of output: Pulsed (or) continuous.
8. Wavelength of the output: 8400Å- 8600Å.
9. Band gap: 1.44eV.

Principle:

The electron in conduction band combines with a hole in the valence band and hence the combination of electron and hole produces energy in the form of light. This photon in turn may induce another electron in the conduction band (CB) to valence band (VB) and thereby stimulate the emission of another photon.

Construction:

The active medium is a p-n junction diode made from a single crystalline material i.e Gallium Arsenide, in which P-region is doped with Germanium and n-region with Tellurium. The thickness of the P-N junction layer is very narrow so that the emitted laser radiation has large divergence. The junctions of the 'p' and 'n' are well polished and are parallel to each other as shown in fig. 3.17

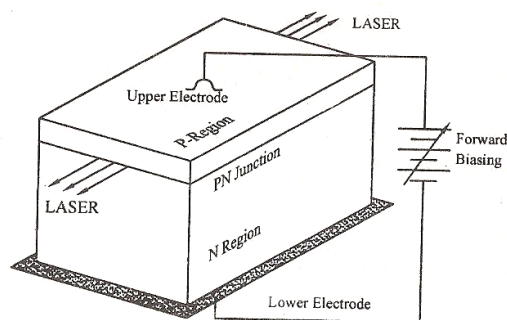


Fig. 3.17 Semiconductor Laser

Since the refractive index of GaAs is high, it acts as optical resonator. So that the external mirrors are not needed. The upper and lower electrodes fixed in the 'p' and 'n' region helps for the flow of current to the diode while biasing.

Working:

1. The population inversion in a p-n junction is achieved by heavily doping 'p' and 'n' materials. So that the Fermi level lies within the conduction band of n type and within the valence band of 'p' type as shown in fig. 3.18

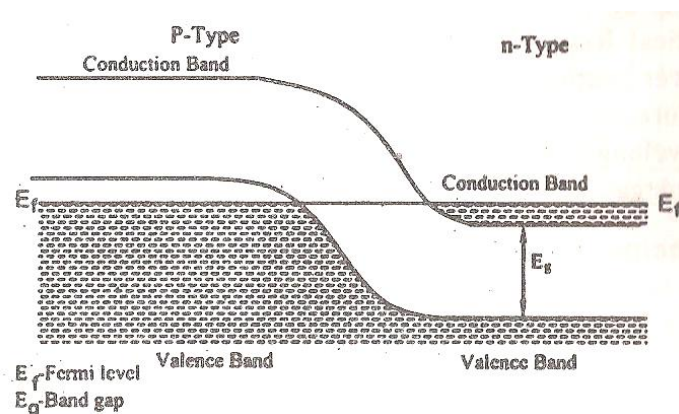


Fig. 3.18

2. If the junction is forward biased with an applied voltage nearly equal to the band gap voltage, direct conduction takes place. Due to high current density active region is generated near the depletion region.
3. At this junction, if a radiation having frequency (ν) is made to incident on the p-n junction then the photon emission is produced as shown in fig 3.19
4. Further, the emitted photons increases the rate of recombination of injected electrons from the n region and holes in p region by inducing more re-combinations.

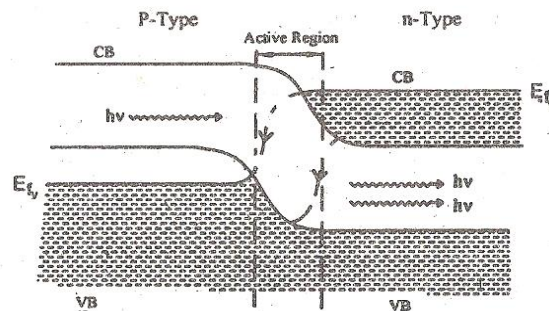


Fig 3.19

5. Hence the emitted photons have the same phase and frequency as that of original inducing photons and will be amplified to get intense beam of laser.
6. The wavelength of the emitted radiation depends on
 - (i) The band gap
 - (ii) The concentration of donor and acceptor atoms in GaAs.

Advantages

1. It is easy to manufacture the diode.
2. The cost is low.

Disadvantages

1. It produces low power output.
2. The output wave is pulsed and will be continuous only for some time.
3. The beam has large divergence

3.11 Application of Laser

(a) Industry

- (i) Welding, cutting, drilling
- (ii) Testing the materials for flaws or defects without damaging them (NDT)
- (iii) Heat treatment of metallic and non – metallic (Plastic, ceramic, glass) materials

(b) Medical

- (i) Treatment of detached retinas
- (ii) Performing micro surgery and bloodless operations

(iii) Treatment of human, animal cancer and skin tumours

(c) Engineering

- (i) Laser beam is used to transmit hundred of messages at a time on radio, television and telephone.
- (ii) Communication between the planets is possible with Laser.
- (iii) It is used in under water communications between submarines.

FIBRE OPTICS

3.12 Introduction:

Soon after the discovery of LASERS some experiments were carried out on propagation of information through light waves in open atmosphere. A light beam acting as a carrier wave is capable of carrying more information than radio waves and microwaves because of its high frequency as shown below

Radio waves 10^6 Hz | Microwaves- 10^{10} Hz | Light Waves- 10^{15} Hz .

Due to atmospheric conditions like rain, fog etc, the efficient communicated systems is disturbed. Hence to have an efficient communication system the light which carries the information requires a guiding media known as **optical fibres**.

Thus the fibres optics is a branch of physics which deals with the transmission and reception of light waves using optical fibres which act as a **guiding media**.

3.13 Optical fibre:

It is made up of transparent dielectrics (SiO_2) (glass or plastics). An optical fibre consists of a central core glass ($50 \mu\text{m}$ diameter) surrounded by a cladding ($125\text{-}200 \mu\text{m}$) which is of slight lower refractive index than core as shown in fig 3.20

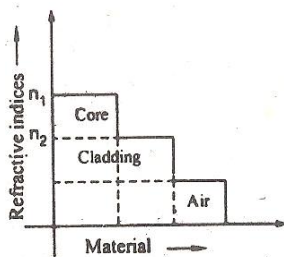


Fig. 3.20

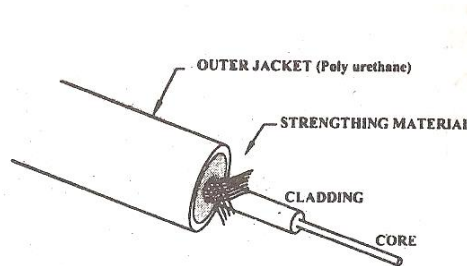


Fig 3.21

The cladding is enclosed by strength members (kevlar) and polyurethane jacket, which acts as protective skin for core and cladding as shown above in the fig 3.21 the protective layer is used so as to make the optical cable to withstand for the need of pulling, bending, scratching, rolling etc. The layer also traps the escaping light from the core.

3.14 Features of optical fibres:

1. It is light in weight.
2. It is smaller in size and flexible, so that it can be bent to any position.
3. It is non-conductive, non-radioactive and non- inductive
4. It has high bandwidth and low loss.
5. There is no short circuiting as in metal wires.

6. There is no internal noise/cross talk.
7. It can withstand for any range of temperature and moisture condition.

3.15 Principle and propagation of light through optical fibres:

For optical fibres, the process of propagation of light is simple, because once the light enters the fibre, the rays do not encounter any new surfaces but repeatedly they hit the same surface. The reason of confining the light beam inside the fibres is the **total internal reflection**.

Even for a bent fibre, the light guidance takes place by multiple total internal reflections all over the length of the fibre as shown in fig 3.22

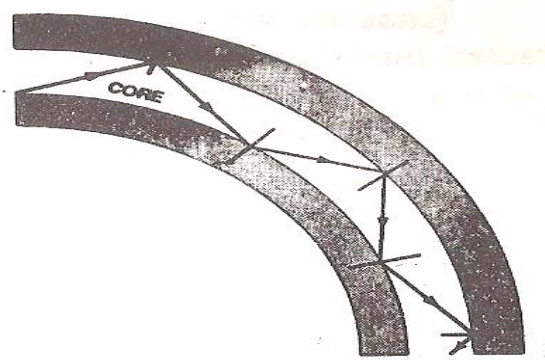


Fig 3.22

Principle: The principle of optical fibre communication is total internal reflection.

Total Internal Reflection:

The phenomenon of total internal reflection takes place when it satisfies the following two conditions.

Condition 1: Light should travel from denser medium to rarer medium

$$\text{i.e. } n_1 > n_2$$

Where n_1 = refractive index of core

n_2 = refractive index of cladding

Condition 2 : The angle of incidence on core should be greater than the critical angle

$$\text{i.e. } \phi > \phi_c$$

Where ϕ - angle of incidence

ϕ_c - Critical angle

Propagation phenomenon:

Let the light ray traverses from denser medium to rarer medium.

Case(i): When $\phi < \phi_c$, the ray traverses along the interface as shown in below figure 3.23

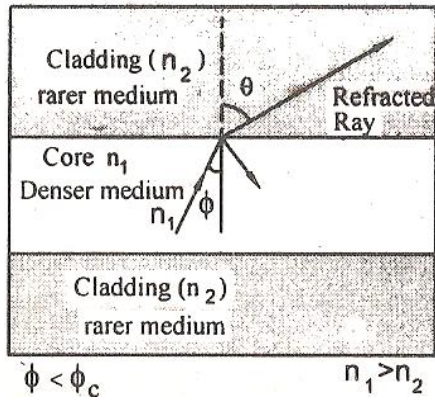


Fig 3.23

Case(ii): When $\phi = \phi_c$, the ray traverses along the interface as shown in below figure 3.24

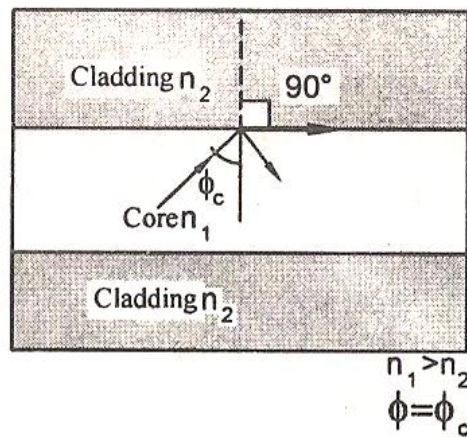


Fig 3.24

So that the angle of refraction is 90° . This angle ϕ_c is called as **critical angle**.

Case (iii) : when $\phi > \phi_c$, the ray is totally reflected back (internally) into the denser medium itself as shown in the below figure 3.25

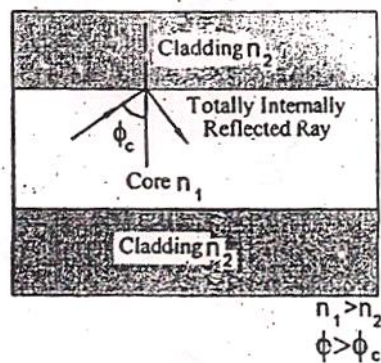


Fig 3.25

From Snell's law (The maximum angle for total internal reflection ϕ_c)

$$n_1 \sin \phi_c = n_2 \sin 90^\circ$$

$$\sin \phi_c = \frac{n_2}{n_1}$$

$$\phi_c = \sin^{-1} \left(\frac{n_2}{n_1} \right)$$

3.16 NUMERICAL APERTURE AND ACCEPTANCE ANGLE:

Let us consider a cylindrical fibre. It consists of core of refractive index n_1 and cladding of refractive index ' n_2 ' and let ' n_0 ' be the refractive index of the medium (air) in which the optical fibre is placed.

The incident ray travel along AO and enters the core at an angle ' i '. The ray is refracted along OB at an angle θ in the core as shown in fig 3.26

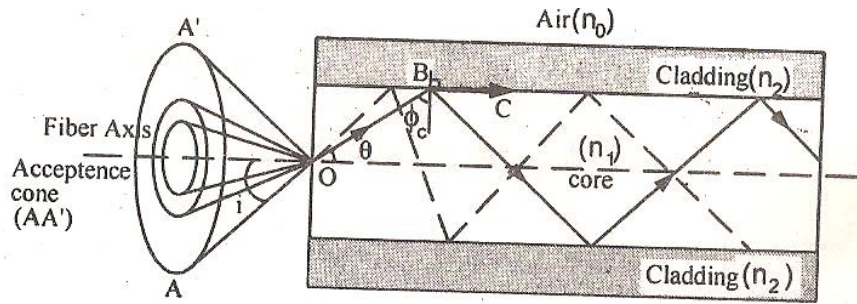


Fig 3.26

It further proceeds to fall at critical angle of incidence $\phi_c = 90 - \theta$ on the interface between core and cladding. At this angle the ray just moves along BC. Any ray which enters into the core at an angle of incidence less than i will have refractive angle less than θ .

Hence the angle of incidence ($\phi = 90 - \theta$) at the interface of core and cladding will be more than the critical angle. Hence the ray is totally internally reflected ray. Thus, only those ray which passes within the acceptance angle (cone) will be totally reflected (internally). Therefore, the light incident on the core within this maximum external incident angle (i_m) can be coupled into the fibre to propagate. This angle (i_m) is called as wave **acceptance angle**.

Expression for Acceptance angle and Numerical aperture:

(i) Applying Snell's law at point of entry of ray (AO) we have

$$n_0 \sin i = n_1 \sin \theta$$

$$\sin i = \frac{n_1}{n_0}$$

$$\sin i = \frac{n_1}{n_0} (\sqrt{1 - \cos^2 \theta}) \dots \dots \dots (1)$$

(ii) Applying Snell's law at point B (I.e. on interface)

$$n_1 \sin \phi = n_2 \sin 90^\circ \dots \dots \dots (2)$$

$$\sin \phi = \frac{n_2}{n_1} \quad (\text{since } \sin 90^\circ = 1)$$

$$\sin(90^\circ - \theta) = \frac{n_2}{n_1}$$

$$\cos \theta = \frac{n_2}{n_1} \dots\dots\dots (3)$$

substituting eq (3) in eq (1) we get

$$\sin i = \frac{n_1}{n_0} \sqrt{1 - \frac{n_2^2}{n_1^2}}$$

$$\sin i = \frac{n_1}{n_0} \cdot \frac{1}{n_1} \sqrt{n_1^2 - n_2^2}$$

$$\sin i = \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$$

$$i = \sin^{-1} \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$$

The angle $i = \sin^{-1} \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$ is the acceptance angle.

If the medium surrounding the fibre is air $n_0=1$

$$i = \sin^{-1} \sqrt{n_1^2 - n_2^2}$$

Numerical Aperture (NA) :

It is defined as sine of acceptance angle.

$$NA = \sin i$$

$$NA = \sqrt{n_1^2 - n_2^2}$$

Acceptance angle:

The maximum angle at or below which the light can suffer **total internal reflection** is called as **acceptance angle**. The cone is referred as acceptance cone

3. 17 TYPES OF OPTICAL FIBERS.

Optical Fibers are classified into 3 major categories based on

- (i) Material
- (ii) Number of modes and
- (iii) Refractive index profile

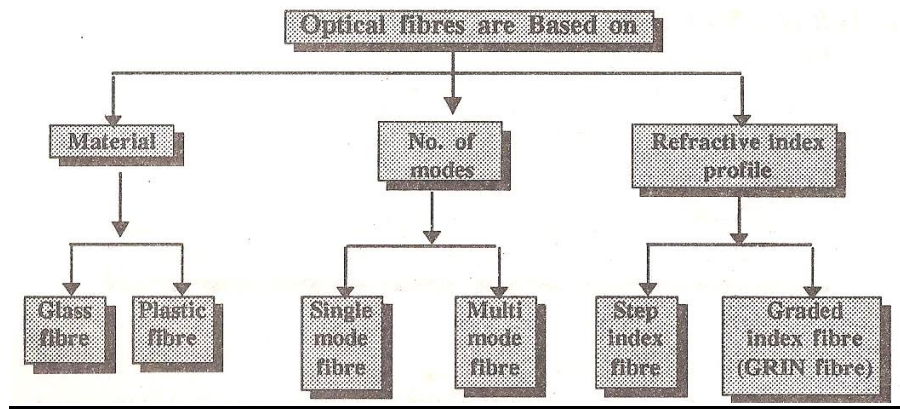


Fig. 3.27 Block diagram for types of optical fiber

3.17.1 Classification based on material:

(i) Glass Fiber:

If the optical fiber is made from fusing mixture of silica glass and metal oxide, it is known as **Glass Optical Fiber**.

Example: Core- GeO_2 - SiO_2 Cladding- SiO_2 .

Advantage:

- Glass fiber optic cable the fastest and the most expensive.
- They can be used for long distances.
- Glass optical fibre exhibit very low losses.

Disadvantage:

- Should be handled with care

(ii) Plastic Fiber

If the fiber is made up of plastics, then it is called **Plastic Fiber**.

Example: Core-Polystyrene Cladding-Methylmethacrylate.

Advantage:

- Cheap, flexible
- Toughness and durability
- Low cost.

Disadvantage:

- high losses

3.17.2 Classification based to Number of modes:

(i) Single Mode Optical Fiber:

1. Single-mode fibers have a narrower core and allow only one, mode of propagation, to be transmitted over very long distances. Single mode fiber carries far more bandwidth, but in a narrower spectral width
2. It allows only one mode to propagate.
3. It has smaller core diameter with difference in refractive index of core and cladding is small
4. No dispersion or degradation of signals during propagation.
5. Information can be carried to longer distance.
6. Launching of light and connecting the fibers are difficult.
7. Fabrication is difficult and costly

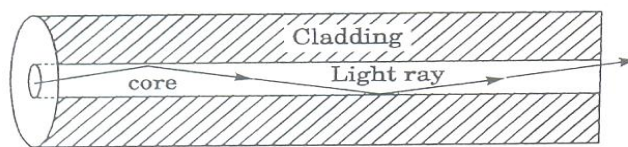


Fig. 3.28 Single mode fiber

(ii) Multi Mode Optical Fiber:

1. Multi-mode fiber has a much thicker core than single mode.
2. It allows a large number of modes of light to propagate through it.
3. It has larger core diameter with difference in refractive index of core and cladding is larger than Single Mode Fiber.
4. There is signal degradation due to multimode dispersion.
5. Information can be carried to shorter distance.
6. Launching of light and connecting the fibers are easy.
7. Fabrication is less difficult and not costly

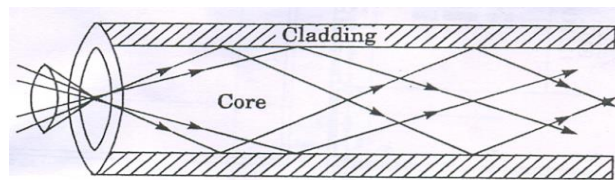


Fig. 3.29 Multimode Fiber

3.17.3 Classification based on refractive index:

Based on the variation in the refractive index of the core and cladding the optical fibers are classified into two types . they are

1. Step –index fibre.
2. Graded –index fiber.

1. Step-index fibre:

In step index fiber, the variation in refractive indices of air, cladding and core vary step by step. Hence this type of fiber is known as **STEP-INDEX** fiber. Based on the refractive index and the number of modes step index fiber is classified into

- Step index- single mode fiber.
- Step index – multi mode fiber.

Step index- single mode fiber:

1. It consists of a thin core of uniform refractive index of a higher value
2. The core is surrounded by a cladding of uniform refractive index lesser than that of the core.
3. Core diameter is 5 to 10 μm external diameter of cladding is 50 to 125 μm .
4. Due to its small core, only a single mode of light ray transmission is possible .
5. Numerical aperture is small.
6. Loss of light is less.
7. Band width is high.

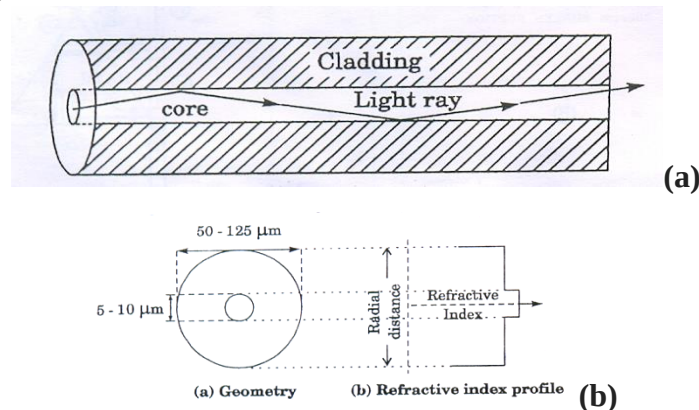
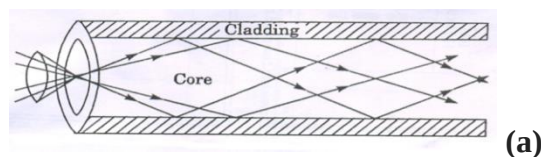


Fig. 3.30 (a & b) Single mode step up index fiber

Step index – multi mode fiber.

1. Its core has much larger diameter which makes it easier to support propagation of a larger number of modes.
2. It has a core material with uniform refractive index and cladding material of lesser refractive index than that of the core.
3. A typical step-index multimode fiber has a core diameter of 50 to 200 μm and external diameter of cladding 125 μm to 300 μm .
4. Because of larger diameter of the core, the propagation of many modes the fiber is possible.
5. Numerical aperture is high.
6. Loss is large.
7. Bandwidth is less



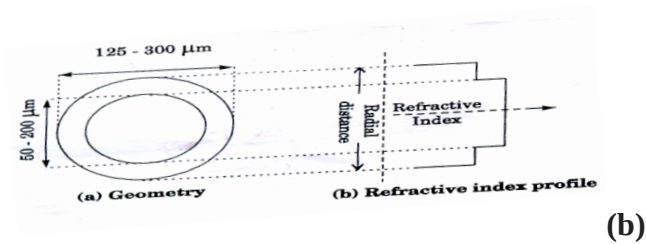


Fig. 3.31 (a&b) Multimode Step up Index Fiber

Graded-index fibre:

1. The refractive index of the core is maximum along the fiber axis and it gradually decreases towards core-cladding interface. Thus it is called as graded index fiber.
2. A typical graded index multimode fiber has a core diameter of 50 to 200μm, and an external diameter of cladding 100 to 250μm.
3. If the core is high, the intermodal dispersion loss must be high.
4. The refractive index of the core is uniform throughout and undergoes step change at the cladding boundary.
5. Attenuation is more for multimode Step Index Fiber but less for single mode Step Index Fiber.
6. Numerical Aperture is more for multimode Step Index Fiber but less for single mode Step Index Fiber.
7. Step index multi mode fibre has a lower band width

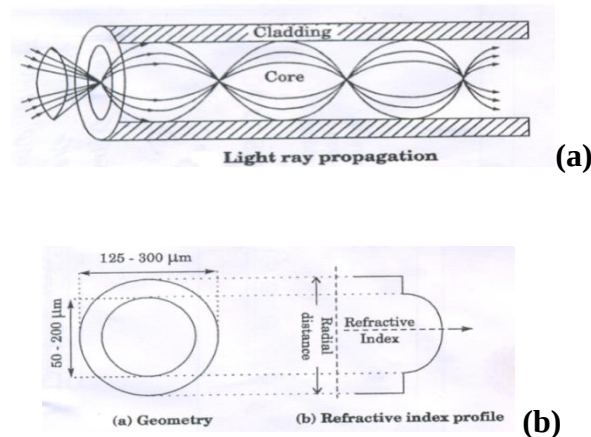


Fig. 3.32 (a&b) Graded Index Fiber

3.18 Differences between step index fibre and graded index fibre:

S.No	Step index fibre	Graded index fibre
1.	The difference in refractive indices is obtained in single step and hence called as step index fibre.	Due to non-uniform refractive indices, the difference in refractive index is obtained gradually from centre towards interface and hence called graded index fibre.
2.	The light ray propagation is in the form of meridional rays and it intersects the fibre axis.	The light propagation is in the form of skew rays and it will not cross the fibre axis.
3.	The path of light propagation is in Zig-Zag manner.	The path of light propagation is helical.

4.	Step index fibre has lower bandwidth.	Graded index fibre has higher bandwidth.
5.	Distortion is more in multimode step index fibre	Distortion is less or practically /no distortion due to self focusing effect.
6.	No. of modes of propagation $N_{\text{step}} = 4.9 \left(\frac{d \times NA}{\lambda} \right)^2 = \frac{V^2}{2}$ Where d = diameter of fibre core λ = Wavelength NA = Numerical Aperture V = V-Number which is less than or equal to 2.405 for single mode fibres and greater than 2.405 for multimode fibres.	No. of modes of propagation $N_{\text{graded}} = \frac{4.9 \left(\frac{d \times NA}{\lambda} \right)^2}{2} = \frac{V^2}{4}$ $N_{\text{graded}} = \frac{N_{\text{step}}}{2}$

3.19. FIBER OPTIC SENSORS

Optical sensors:

Optical sensors are a transducer which converts any form of signal into optical signal in the measurable form.

Types:

Intrinsic sensor (Active sensor): Example: Pressure sensor, Temperature sensor, Liquid level sensor.

Extrinsic sensor (Passive sensor): Example: Displacement sensor.

(i) Active sensor:

In the case of intrinsic (or) active sensors physical properties to be sensed acts directly on the fiber itself to produce change in transmission characteristics. Eg: phase sensor, Temperature sensor, pressure sensor.

(ii) Passive sensor

These types of sensor have separate sensing element and the fibre acts only as wave guiding media, in this type of sensor the fibre is used largely as transmission medium. Eg: displacement sensor

3.20 Temperature sensor:

Principle:

The absorption of light by the silicon layer varies with temperature and the variation modulates the intensity of the light received at the detector

Description:

It consists of a light source, a detector, two fibre (transmitting fibre, receiving fibre) and silicon covered with a reflecting coating as shown in fig. 3.33

Working:

1. When the light is passed through the fibre from the source and the light after reflecting from the silicon layer returns to a detector. During this process, light is absorbed by the silicon layer.

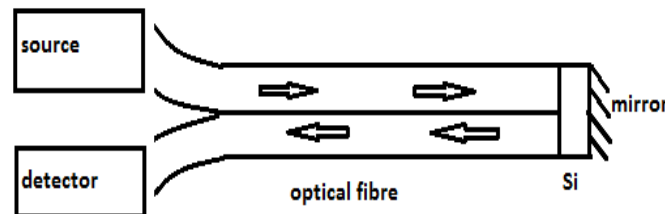


Fig. 3.33 Temperature Sensor

2. The absorption of light by silicon layer varies with temperature and this variation modulates the intensity of the light received at the detector.

3. A photo detector is used to measure the intensity of light and temperature is sensed. Temperature in the range from 80°C to 700°C is measured using this technique.

3.21 Displacement sensor:**Principle:**

It is working on the principle that the intensity of light signal is modulated varying the distance between fibre cable and object.

Description:

It consists of two group of fibre optics. One set is connected to a light source is called as **transmitting fibre**. The other set is connected to a photo detector and is known as the **receiving fibre**. The displacement sensor is shown in fig.3.34.

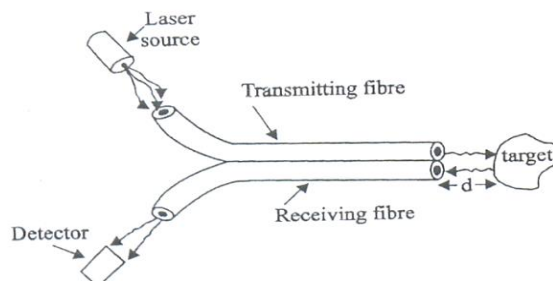


Fig. 3.34 Displacement Sensor

Working:

1. Light generated from the source is passed through the transmitting fibre.
2. Then the light travels to the target surface and part of it is reflected back to the receiving fibre.
3. This reflected light is transmitted to photo detector where intensity is measured.
4. The intensity of light is function of distance between receiving fibre and moving object.
5. A calibration graph is drawn between displacement and intensity

3.22 Fibre optic communication system:

A fiber optic communication system is very much similar to a traditional communication system. Its purpose is to transfer an information from a source to a distant user.

Principle:

The basic principle of optic fibre communication is transmission of information over a required distance by the propagation of optical signal through optical fibre.

Description:

The main parts of fibre optical communication system are,

1. Information signal source
2. Transmitter
3. Light source
4. Propagation medium
5. Receiver

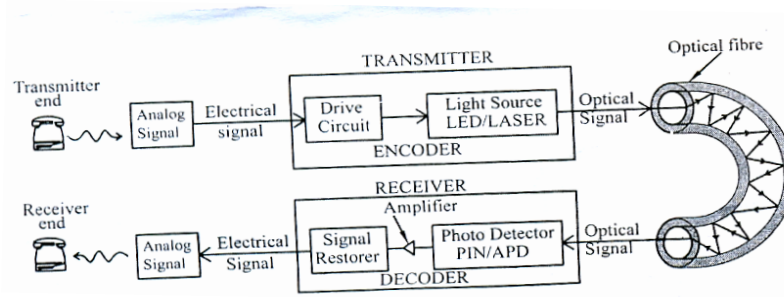


Fig. 3.35 Schematic diagram of Fiber optic communication system

1. Information signal source:

The information signal source may be voice, music, digital data or analog voltage and video signal. Here it is analog information.

2. Transmitter:

It consists of a drive circuit and a light source. The drive circuit transfers the electric input signals (information signal) into digital pulses.

3. Light source:

The light source generates optical energy which acts as the information carrier. Laser or LED is used for light sources. This light source converts digital pulses into optical pulses.

4. Propagation medium

The optical fibre is used as a propagation medium. It acts as a waveguide and transmits optical pulses towards receiver by the principle of total internal reflection.

5. Receiver:

It consists of a photo detector, an amplifier and a signal – restoring circuit. The photo detector detects optical signal and converts back into an electrical signal. The signal is amplified and it converts from digital to analog signals.

Working:

Analog information such as the voice of a telephone user produces electrical signals in analog form this analog electrical signal is converted into digital signal which is in the form of electrical pulses. The electrical signal modulates the light emitted by an optical source (such as a LED or laser diode). Now, this optical signal is fed into the fibre.

At the receiving end, the optical signal passing through the fibre is fed into a photo detector. The photo detector detects optical signal and converts it into pulses of electric current. This digital signal is once again converted into analog signal. This analog signal contains the same information as transferred from transmitting end. By this way, the information is transmitted from one end to other end.

3.23. Benefits of optical fiber communication system:

- a) Immense bandwidth to utilize and very low transmission loss,
- b) Total electrical isolation in the transmission medium
- c) Small size and light weight,
- d) High signal security & Immunity to interference and crosstalk
- e) Very low power consumption and wide scope of system expansion etc.

3.24. Application of optical fibres :

(a) Industries

- (i) Fiber optic endoscope are used for finding minor cracks, pores etc in big machines.
- (ii) They are used in coagulation in chemical industries and laboratories
- (iii) It is also used to connect a monitoring station and remote sensor in industries
- (iv) Optical fibers are used as a sensor in the measurement of pressure, temperature, absence (or) presence of an object, used as a link between a remote sensor and a monitoring station.

(b) Medical field

- (i) Fiber optic endoscope are used in medical diagnosis
- (ii) It is used to visualize the inner organs of the body.

- (iii) Fibres as endoscopes are used in various medical fields such as cario scopy, laparoscopy, cystocopy etc.,
- (iv) They can be made in extremely thin, flexible strands for insertion into the blood vessels, lungs, and other hollow parts of the body. Depending on the field in which they are used. It is a tubular optical instrument, used to inspect the internal parts of a human body which are not visible to the naked eye.

(c) Communications

- a) Long haul communication & Transmission of TV signals
- b) Large numbers of channels are handled for communication

(d) Computer networks

LAN (Local Area Network), connecting the computer with printer, workstation, multi-beam radars.

3.25. Application of fibre optic sensors :-

1. It is used as liquid level indicator based on the principle of change in refractive index of the medium.
2. Optical displacement sensors are used to find the position and displacement of the target.
3. Temperature sensors are used to measure the temperature accurately.
4. The pressure can accurately measured using photo elastic pressure sensors.
5. Laser Doppler velocimeter (LDV) is used to measure several physical quantities such as velocity = fluid surface velocity.

Pondicherry University 2 marks Question and Answer

1. What is meant by population inversion?(April 2011, May 2012 & 2013)

The process of achieving large population of atoms in the upper level ($N_2 > N_1$) than the lower level is said to be population inversion.

2. What is called stimulated emission?(Jan 2013)

An atom in the ground state with energy E_1 absorbs a photon of energy $h\nu$ and goes to the excited state with energy E_2 provided that the photon energy $h\nu$ is equal to the energy difference ($E_2 - E_1$). This process is called stimulated emission.

3. Define spontaneous emission and stimulated emission.(Jan 2014)

(i) Spontaneous emission:

The atom in the excited state returns to the ground state by emitting a photon of energy $E = E_2 - E_1 = h\nu$, spontaneously without any external triggering. This process is known as spontaneous emission.

(ii) Stimulated emission:

The atom in the excited state can also return to the ground state by external triggering (or) inducement of photon thereby emitting a photon of energy equal to the energy of the incident photon, known as stimulated emission.

4. Explain the classification of optical fibres based on the materials? (Jan 2013)

Based on material: glass fibre, plastic fibre

5. What are the characteristics features of a laser beam? (Jan 2010)

The characteristics features of a laser beam are perfect monochromatic, Extremely intense, High Directionality, Highly coherent.

6. Define total internal reflection? (May 2013, Jan 2016)

The phenomenon of total internal reflection takes place when it satisfies the following two conditions.

Condition 1: Light should travel from denser medium to rarer medium

$$\text{i.e. } n_1 > n_2$$

Where n_1 = refractive index of core

n_2 = refractive index of cladding

Condition 2 : The angle of incidence on core should be greater than the critical angle

$$\text{i.e. } \phi > \phi_c$$

Where ϕ - angle of incidence

ϕ_c - Critical angle

7. Explain temporal coherence. (May 2011)

The maximum length of the resulting wave train on which any two points can be correlated is called temporal coherence.

8. What is the basic principle of fibre optics? (May 11, 17, Jan 2016)

Total internal reflection is the principle behind the transmission of light waves in an optical fibre.

9. Difference between spontaneous and stimulated emission(May 2017)

S.No	Spontaneous Emission	Stimulated Emission
1.	Emitted Photon travels in random direction	Emitted Photon can be made to travel in same direction.
2.	Emitted Photon cannot be controlled.	Emitted Photon can be controlled.
3.	Emitted Photon is less intense and is incoherent.	Emitted Photon is highly intense and is coherent.
4.	Emission of light radiation is not triggered by any external influence	Forced emission of light radiation caused by incident photons.

10. What are the different types of pumping in Lasers?(May 2017)

Different Types of pumping are

- Optical pumping,
- Direct electron pumping,
- Inelastic atom-atom collisions and
- Chemical reaction.

Additional 2 marks Question and Answer

Part - A

1. What does the term Laser stand for?

Laser is an acronym for Light Amplification by Stimulated Emission of Radiation.

2. Explain the principle of laser

An atom that has only two energy levels, E_1 and E_2 , when exposed to radiation having a stream of photons, each with energy $h\nu$, three distinct processes can take place. They are absorption, spontaneous emission and stimulated emission. If the stimulated emission is prominent then laser action will take place.

3. What are the conditions needed for laser action?

- (i) The rate of absorption should be equal to rate of emission.
- (ii) Stimulated emission should be prominent over spontaneous emission and absorption.
- (iii) Number atoms in the excited state must be higher than the ground state.
- (iv) Losses in the lasing medium should be overcome by induced transition in the amplifying medium.

4. What is meant by pumping of atoms and mention types?

The excitation of atoms in order to achieve the population inversion can be done by the help of external energy is referred as pumping. Types are

Optical pumping, direct electron pumping, Inelastic atom-atom collisions, Chemical reaction.

5. What are the different types of laser?

Based on the active medium, they are classified as Solid state lasers, Gas lasers, Liquid Lasers, Dye lasers and Semiconductor lasers.

6. Mention any two applications of laser.

Laser beams are used to obtain a three dimensional lensless holograms.

Laser is used in printers for computer printouts.

7. Give the significant characteristics of laser?

Directionality, Monochromaticity, High intensity, Coherence.

8. Write the relation between Einstein Coefficients and give the details.

$B_{12} = B_{21}$, The probability of absorption per unit is equal to the probability of stimulated emission and $A_{21} / B_{21} = 8\pi h\nu^3 / c^3$, The ratio of spontaneous emission to stimulated emission is directly proportional to frequency of the radiation.

9. What are the uses of optical resonator in Laser? (Jan 2016)

The optical resonator, or optical cavity, is two parallel mirrors with optical coating placed around the active medium which provides feedback of the light for amplification. Typically one will

be a high reflector, and the other will be a partial reflector. The partial reflector is called the output coupler, because it allows some of the light to leave the cavity to produce the laser's output beam.

10. What is the basic principle of fibre optics?

Total internal reflection is the principle behind the transmission of light waves in an optical fibre.

11. State the properties that an optical fiber should possess

- i. Efficient guide for the light waves
- ii. Low scattering cross section
- iii. The absorption, attenuation and dispersion of optical energy are low.

12. Give the classification of optical fibers based on material, mode and refractive index. (Jan 2014, 2016)

Based on modes: single mode fibre, multimode fibre

Based on material: glass fibre, plastic fibre

Based on refractive index: step-index fibre, graded-index fibre.

13. Define acceptance angle

The rays which are incident on fibre core within conical half angle ϕ_2 will be refracted into fibre core and hence propagate into the core by total internal reflection. This angle is referred to as acceptance angle.

PROBLEMS :

1. A step index fiber has a numerical aperture of 0.16 and a core refractive index of 1.450. Calculate the refractive index of the cladding.

$$\text{Numerical aperture} = \sqrt{(n_1^2 - n_2^2)}$$

$$0.0256 = 1.45^2 - n_2^2$$

$$n_2^2 = 2.0769$$

$$n_2 = 1.441$$

2. A fibre has a core diameter of 6 μm and its core refractive index is 1.45. The refractive index of the cladding is 1.448. Determine the maximum number of modes allowed to propagate through the fiber, if the wavelength of light is 1 μm .

Given data

Refractive index of core (n_1) = 1.45

Refractive index of cladding (n_2) = 1.448

Diameter of the core (d) = 6 $\times 10^{-6}\text{m}$

Wavelength of light = 1 $\times 10^{-6}\text{m}$

$$\begin{aligned} \text{Solution : Numerical aperture} &= \sqrt{(n_1^2 - n_2^2)} \\ &= \sqrt{(1.45^2 - 1.448^2)} = 0.07613 \end{aligned}$$

No. of modes of propagation through the fiber

$$N_{\text{step}} = 4.9 \left(\frac{d \times NA}{\lambda} \right)^2 = 4.9 \left(\frac{6 \times 10^{-6} \times 0.07613}{1 \times 10^{-6}} \right)^2 = 1.022$$

Since $N=1$ i.e allowing only one mode, it is a single mode fibre.

3. Find the numerical aperture and acceptance angle for a step-index for which $n_1=1.5$, $n_2=1.49$ and $n_0=1$

Solution:

Refractive index of core is $n_1=1.5$

Refractive index of cladding is $n_2=1.49$

Numerical aperture (NA) = $\sqrt{(n_1^2 - n_2^2)} = \sqrt{(1.5^2 - 1.49^2)} = 0.1729$

Acceptance angle $\sin i_m = NA$

$i_m = \sin^{-1}(NA) = 9.956474 = 10^\circ 35'$

4. A fiber has diameter of $6\mu\text{m}$ and its core refractive index is 1.47 and for cladding it is 1.43. How many modes can propagate into the fiber if the wavelength of the laser source is $1.5\mu\text{m}$.

Solution:

$NA = \sqrt{(1.47^2 - 1.43^2)} = 0.34$

$$N_{\text{step}} = 4.9 \left(\frac{d \times NA}{\lambda} \right)^2 = 4.9 \left(\frac{6 \times 10^{-6} \times 0.34}{1.5 \times 10^{-6}} \right)^2 = 9.06 = 9 \text{ modes}$$

5. Find the numerical aperture and acceptance angle of an optical where the refractive index of the core is 1.54 and the cladding is 1.5.

Solution: Refractive index of core is $n_1 = 1.54$

Refractive index of cladding is $n_2 = 1.5$

(i) Numerical aperture (NA) = $\sqrt{(n_1^2 - n_2^2)} = \sqrt{(1.54^2 - 1.5^2)} = \sqrt{0.1216} = 0.3487$

(ii) Acceptance angle = $\sin^{-1}(NA) = \sin^{-1}(0.3487) = 20^\circ 40'$

6. What is the ratio of the stimulated emission to spontaneous emission at a Temperature of 250°C for the sodium D line? (Given data $\lambda = 5.9 \times 10^{-7}\text{m}$, $h = 6.626 \times 10^{-34}\text{Js}$) (April 2011)

$$T = 250^\circ\text{C} = 250 + 273 = 523$$

$$\frac{\text{Stimulated emission}}{\text{spontaneous emission}} = \frac{R_{21}(\text{ST})}{R_{21}(\text{SP})} = \frac{1}{e^{\left(\frac{h\nu}{k_B T}\right)} - 1}$$

$$\frac{\text{Stimulated emission}}{\text{spontaneous emission}} = \frac{R_{21}(\text{ST})}{R_{21}(\text{SP})} = \frac{1}{e^{\left(\frac{hc}{k_B \lambda T}\right)} - 1}$$

$$\frac{\text{Stimulated emission}}{\text{spontaneous emission}} = \frac{R_{21}(\text{ST})}{R_{21}(\text{SP})} = \frac{1}{e^{\left(\frac{6.625 \times 10^{-34} \times 3 \times 10^8}{1.38 \times 10^{-23} \times 523 \times 5890 \times 10^{-10}}\right)} - 1}$$

$$\frac{\text{Stimulated emission}}{\text{spontaneous emission}} = \frac{R_{21}(\text{ST})}{R_{21}(\text{SP})} = 5.329 \times 10^{-21}$$

7. Calculate the angle of acceptance and critical angle of an optical fiber whose refractive indices of the core and the cladding are 1.563 and 1.49 respectively (Jan 2016).

Solution: Refractive index of core is $n_1 = 1.563$

Refractive index of cladding is $n_2 = 1.49$

(i) Numerical aperture (NA) $= \sqrt{(n_1^2 - n_2^2)} = \sqrt{(1.563^2 - 1.49^2)} = 0.4721$

(iii) Acceptance angle $= \sin^{-1}(\text{NA}) = \sin^{-1}(0.4721) = 28^\circ 53'$

(iv) Critical angle $\phi_c = \sin^{-1}[n_2/n_1] = \sin^{-1}[1.49/1.563] = 72^\circ 41'$

8. Calculate the wavelength of emission from GaAs whose band gap is 1.44eV. (Jan 2016).

Solution: Bandgap of given GaAs semiconductor (E_g) = 1.44eV

$$= 1.44 \times 1.609 \times 10^{-19}$$

$$= 2.3 \times 10^{-19} \text{ J}$$

$$\text{w.k.t } \lambda = hc / E_g = 6.626 \times 10^{-34} \times 3 \times 10^8 / 2.3 \times 10^{-19} = 8.6 \times 10^{-7} \text{ m} = 0.86 \mu\text{m}$$

Part B Important Questions

1. Explain the principle and working of a semiconductor laser (Jan 2016)
Answer : Refer Page No. 17 to 20
2. Explain the construction and working of a Nd-YAG laser (April 2011, May 2013)
Answer : Refer Page No. 12 to 14
3. Discuss the construction and working of CO₂ laser (Jan 2011, 2014, May 2012, 2017)
Answer : Refer Page No. 14 to 17
4. Explain the principle and action of laser with suitable energy level diagram.
Answer : Refer Page No. 8
5. Using Einstein's theory of stimulated absorption, spontaneous emission and stimulated emission. Obtain the relationship between Einstein coefficients A and B. (April 2011, Jan 2016)
Answer : Refer Page No. 4 to 8
6. What is meant by acceptance angle for an optical fiber? Show how it is related to numerical aperture? (May 2012, 2013, Jan 2016)
Answer : Refer Page No. 24 to 25
7. Distinguish between spontaneous emission and stimulated emission
Answer : Refer Page No. 36
8. Explain the types of fiber on the basis of material, number of mode and refractive index.
Answer : Refer Page No. 25 to 29
9. Give any two application of an optical fibre.
Answer : Refer Page No. 33 to 34
10. Derive an expression for the acceptance angle and numerical aperture of an optical fibre. (Jan 2013)(Jan 2014)
Answer : Refer Page No. 24 to 25
11. What is population inversion? Explain why laser action cannot occur without population inversion between atomic levels? (May 2011)
Answer : Refer Page No. 9
12. . Explain (May 2011, 2017)
 - (i) Step index fibre and
Answer : Refer Page No. 29 to 30
 - (ii) Graded index fibre
Answer : Refer Page No. 29 to 30
13. Write an essay on the preparation, advantages of optical fiber communication (May 2017)
Answer : Refer Page No. 32 to 34
14. Describe the propagation of light through an optical fiber (May 2017)
Answer : Refer Page No. 22 to 24